

Machine learning based effective linear regression model for TSV layer assignment in 3DIC

K. Pandiaraj^{*}, P. Sivakumar, K. Jeya Prakash

Department of Electronics and Communication Engineering, Kalasalingam Academy of Research and Education, Krishnankoil, Srivilliputtur, Tamilnadu 626126, India

ARTICLE INFO

Keywords:

Tsv
through silicon via
Ic- integrated circuit
Elrm- efficient linear regression model. ml-
machine learning

ABSTRACT

On the integration of 3D IC design, thermal management issues play a significant role. So, it is required to implement an effective approaches and solutions for integrating 3DIC. The TSV causes problems with the distinct coefficients of thermal expansion that induces mismatch strains and stresses. The major drawback of 3DIC is the thermal management issues which increases the power consumption through the current crowding, perhaps the temperature upraised by the slacked layers due to its heat generation. Several research has not been undergone in 3DIC utilizing machine learning approaches which is highly complicated. This paper firstly proposes an efficient ML model to achieve better reduction in wire length and temperature. An efficient linear regression model is preferred here in order to achieve significant performances in TSV layer assignment. The linear regression utilized gradient based approach where the error is predicted at every instance through tracing gradient cost function. An optimized TSV layer assignment is achieved with this flexible ELRM. The performance analysis of data shows that the proposed ELRM based TSV assignment achieved better wire length and temperature. The ISPD98 Circuit Benchmark Suite is utilized for result evaluation and it achieves improved TSV layer assignment through reducing wire length and temperature.

1. Introduction

A net is a set of two or more cell pins that have the same electrical potential in the final chip design. The circuit net list includes all the nets in the design. During the routing stage, all terminals of the nets in the circuit net list must be properly connected while respecting constraints (e.g., design rules, routing resource capacities) and optimizing routing objectives like minimum total wire length, maximum timing slack, including avoiding blockages due to areas occupied by thermal vias, incorporating the effect of temperature on the delays of the routed wires.

3D stacking is the unique evolving technology where the multiple dies are stacked together in vertical manner for accomplishing greater transistor density without considering device scaling. The interconnect power and delay is reduced providing high density vertical interconnects. Furthermore, in 3-D ICs different circuit technologies are integrated into a single chip, thus new-fangled system-on-chip architectures are revealed that is not existing in 2-D technology. There is an outcome of significant performance, new thermal, signal integrity, power delivery, reliability and novel architectural prospects whilst 3-D integration occurs. The new architectural models and approaches for

design space exploration (DSE) are required for the CPU design space expansion [1]. In order to address the specific 3DIC design challenges, new design tools and approaches are essential.

As the interconnection intricacy increases, the research importance on global routing is also enhanced in search for rapid search paths reporting estimation of wire-length to placer or high-routability results. An evolution-based rip-up and adaptive pseudorandom net-ordering routing was utilized as routing mechanisms. In order to facilitate the dynamic programming based layer assignment, this two efficient via-minimization method was implemented. It was not probable to have less overflows and fast convergence with the use of only one cost function. So after the initial completion of routing for most of nets, left over hard to route nets was handled with one more cost function. This ideal approach was employed to minimize the vias as well.

Thus, the high routing speed, shorter wavelength and lesser runtime was accomplished with this approach. First of all open defects On TSVs should be analyzed correctly to find its defects [2]. Hence author recommended two characterization method for estimating the delay occurred by means of rerouting based on spare TSVs and resistive open defect. The optimization method was also presented based on the integer

^{*} Corresponding author.

E-mail addresses: pandiaraj@klu.ac.in (K. Pandiaraj), siva@klu.ac.in (P. Sivakumar), k.jeyaparakash@klu.ac.in (K.J. Prakash).

<https://doi.org/10.1016/j.micpro.2021.103953>

Received 25 November 2020; Received in revised form 21 December 2020; Accepted 7 January 2021

Available online 10 January 2021

0141-9331/© 2021 Elsevier B.V. All rights reserved.

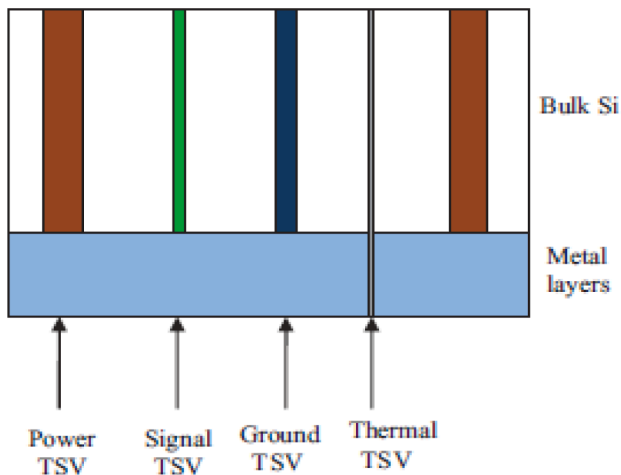


Fig. 1. General TSV.

linear programming (ILP) through allocating the spares to practical TSV [3]. In such a way that the functional TSV spare was neither kept too closer nor far away for ensuring the extra delay caused by the rerouting under upper limit [4]. In 3D integrated circuits (ICs) with through-silicon vias (TSVs) offer new levels of efficiency, power, performance, and form-factor advantages to the semiconductor industry. The different types of TSV Representations are given in Figure 1

In this work, an Improved Gradient Based Linear Regression Algorithm is proposed to obtain better optimization of 3DIC method in terms of reducing the wire length and temperature. To determine whether the machine learning linear regression algorithm fits for layer assignment to attain effective TSV optimization. An effective TSV optimization could be achieved through avoiding thermal mismatch. The linear regression model comprising gradient descent achieves minimized error at each training instance through tracking the cost function of gradient, the fitness function is also calculated to achieve the effective TSV layer assignment. Thus, the effective linear regression is determined with reduction of temperature rise and the wire length and temperature of 3DIC TSV layer is reduced as much as expected.

The main contribution of this work is given below

- To implement Improved Linear regression Algorithm for the better optimization of 3DIC method.
- To minimize Thermal and Wire length for TSV based 3D IC.
- To analyze the performance of the Optimization technique

The comprehensive description of the existing works of 3D IC technology is mentioned in section II. The working procedure of proposed work is briefly described in Section III. The performance, verification of the proposed work is discussed in the Section IV. At last the research work is concluded.

2. Related works

Though 3D IC design is a challenging task, most of the research has not been undertaken due to its complexity in the placement of 3D IC. To augment the efficiency of 3D IC placement, it is required to exploit an effectual die-die thermal coupling. So the Athikulwongse suggested two effective heuristics named as thermal coupling-aware placement (CA), TSV spread and alignment method (TSA) [5]. In this approach, a force directed thermal aware placement was employed to attain a shorter routed wire length. The TSVs were distributed on every single die and it was vertically aligned to accomplish the reduction in local power density and also for increasing the thermal conductivity. Also, the high power logic cells were relocated to the heat sink possessing higher conductivity in order to vertically overlap the high power cells below

TSVs. The impact of TSVs over the 3DIC temperature was evaluated in this framework in terms of analyzing power and thermal analysis of 3D IC design. The multiple dies in 3DIC could be incorporated efficiently into single 3D stack using TSV technology. Because of sheer size of TSV, it occupied significant silicon area which takes larger impact for the quality of 3D IC. The design scheme implemented in this approach was [6] TSV co-placement and TSV site and associated procedure for finding optimized locations of TSVs and gates. The routing congestion could be improved with the well managed TSV, however ill managed TSVs rises the wire length and die area. In order to facilitate unvarying TSV, two TSV assignment algorithms were elucidated. The analysis of the simulation result shows that the 3D IC wire length was smaller than wire length of 2D IC.

The multiple clock domain rises issues in both power and integrity due to its similar sharing of 3-D global PDN. Here the power supply noise distribution on every single tier was examined firstly to analyze the situations leading most awful noise. Furthermore, thermal analysis and the heat distribution among tiers and clock domain was also examined [7]. The frequency and clock domain structure that influence the power supply noise was also enumerated along with temperature distribution. Therefore, the analysis of the experiments demonstrated that the multiclock domain persuade an extreme noise, through-silicon vias could subsidize the heat transfer among tiers and power supply noise. Still there is a lack of research in design portion of 3DIC in terms of quality, it is required to focus on implementation of the new-fangled design which enhances the quality and the runtime. The author suggested an efficient Stochastic Evolution algorithm to resolve the R&R problem with simple routing method (such as maze routing). For this purpose, distinct functions of StocE algorithm was offered. This stocE algorithm was used to control the [8] net selection and tuning of R&R process parameters. The suggested global router acquired lesser components than the conventional methods. The evaluation was made on the ISPD2008 and ISPD98 benchmarks

The experimental results were proved with the analysis of effective delay analysis. Similar study has been articulated in this work [9] where the optimal spare problem in TSV assignment was formulated for uneven TSV placement. The main motive of this work was to minimize total area overhead and the timing constraints. To address these issues, two efficient heuristics was developed. The area overhead persuaded via fault tolerance mechanisms could be reduced using this heuristic including spare TSVs, control logics rerouting wires. The novel framework was implemented in this paper [10] in order to create the 3D NoCs which encounter the constraints of application performance that reducing power dissipation. For the integration of application specific 3D networks on chip, a novel MORPHEUS framework has been introduced which integrates thermal-aware core layout, 3D topology and route generation, serialized vertical through silicon vias and placement of network interfaces and routers. Thus, the solution generated in our framework attained less power dissipation and average latency over 2D NoCs and 3D NoCs as well. Due to higher demand of 3D IC, the author focused the current and future contests of 3-D IC design. TSV-based 3-D IC was the primary focus of this work comprising recent advances in monolithic 3-D ICs. The 3D IC design challenges was discussed concerning the tools and methods. Additionally, the co-design methodology was also deliberated for addressing more interdependent and complicated design problems [11]. Similarly, overwhelmed schemes on tasks were discussed to make commercially feasible 3-D IC technology. In the following paper the author discussed the thermal challenges which is one of the significant roles in 3D IC. TSV thermal stress-aware floor plan technique was introduced aimed at decreasing the thermal load of TSV while designing floor plan. [12] The thermal cycling pattern and thermal gradients were also considered as significant part in this work which persuades the mechanical reliability in run time. The crack occurrence on thinned silicon substrate could be reduced through controlling the temperature gradients and eradicating the damaged thermal cycling patterns. Thus, the design time and run time management scheme was

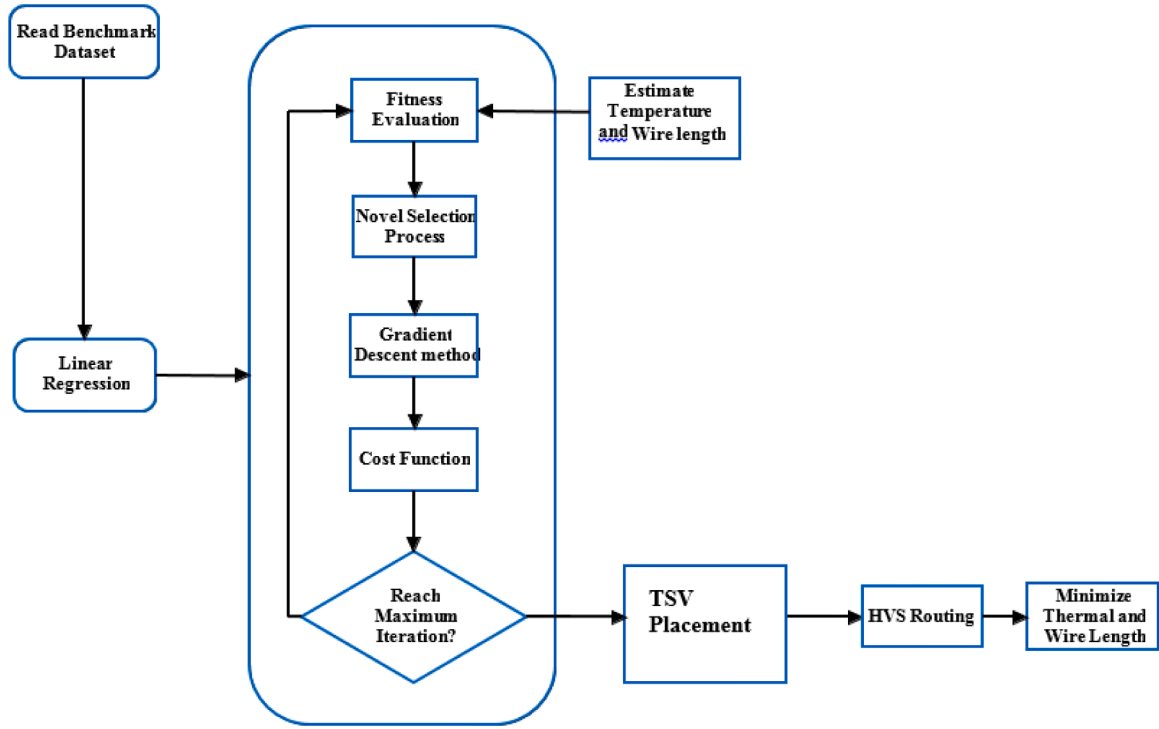


Fig. 2. Workflow of Proposed Methodology.

successfully employed for lessening the mechanical stress on chip. The current crowding is also an important factor affecting the TSVs and the associates among TSVs and power wires, so it must also be considered to analyze thorough current density distribution inside TSVs efficiently. [13]. Thus, from the simulation result it was proved that the PSIM was efficient in terms of identifying high current density and IR drop across the PDN which facilitates designer for better optimization of 3D IC. The design of 3D power supply networks was studied to improve the IR-drop and dynamic noise by implementing the new techniques. This work analyzed the distributed power among the 1000-core processor using 30 stacked dies and the scaling trends those concerning augmented power distribution TSVs and stacking together [14][25]. For the further improvement in minimizing IR-drop and dynamic noise, several techniques were also examined lastly. The monolithic 3D integration of memory and logic over the random vertical stacking order was demonstrated including the capability of using traditional [15]

Inter-layer vias for connecting any 3D IC layer among them. The performance gain of 3D-FPGA was evaluated, and its compatibility and scalability of proposed 3D FPGA was superior to RRAM technology which clearly depicts its greater potentiality. The benefit of monolithically stacked 3-D FPGA was deliberated in this work [16] and 3D NoC design optimization was discussed in the paper [17] to acquire an efficient energy with machine learning approaches. The author in this paper made prior analysis of electrical characteristic of TSV to capture the complex structural defects such as pinholes and void that defects the TSV parameters [18][24]. So, it was necessary to implement new designing techniques for testing the interconnects of TSV. For this purpose, a novel nondestructive defect detection method was implemented using machine learning (ML). The supervised ML approach was utilized. To form a classification model from training S-parameter data sets covering the usual and defected TSV, a supervised ML approach is used. The random forest classifier performance was validated for numerous quantity of void, short, and TSV-based 3-D-stacked ICs open defects with agreeable results [19]. The limitations over the design and run time of 3DIC is discussed in this literature section to analyze the research challenges of 3D IC. Some of the significant challenges and its overcoming strategies for 3D ID TSV placement is given below

- 1 The current density distribution should be properly observed to reduce the thermal rise due to its major impact over TSV.
- 2 Utilizing the heuristics such as spare TSVs and control logics rerouting wires, the area overhead influenced via fault tolerance mechanisms could be abridged.

3. Proposed work

The 3DIC based TSV approach of proposed method is deliberated below in Fig 2 where the Modified Stochastic based Genetic Algorithm is utilized. The proposed method flow diagram is given below and the techniques utilized for effective linear regression is also described. In this approach, gradient descent computation and fitness function for detecting the node is performed. Despite the fact calculating the fitness, the thermal and wire length is assessed for the TSV random placement. Then in the selection process, TSV position is selected by updating the chromosome positions. After this cost function is calculated along with the movement of TSV based on the TSV thermal condition. The wire length and temperature is highly reduced with the use of novel linear regression model.

3.1. Effectual linear regression model

Many machine learning algorithms uses linear regression as an important building block where the models are produced from a specified training set of sample data points through its appropriate linear curve. The real valued function of the linear model over d dimension is properly in the form of $p \rightarrow (\theta, p)$, Here input vector is depicted as $p \in L^d$, $\theta \in L^d$ is the parameters vector that specified the model, Among θ and x the inner product is given as $(\theta, x) = \sum_{k=1}^d \theta_k p_k$. The function (θ, x) used for predicting the output having the features with linear combination signified in x .

Learning algorithm takes access towards training set samples which represents the relation of input –output the model trying to replicate. These are represented as $(x^{(1)}, y^{(1)}), \dots, (x^{(m)}, y^{(m)})$ with $x^{(i)} = (x_1^{(i)}, \dots, x_d^{(i)}) \in L^d$ and $y^i \in L$ for $i \in [m]$.

The linear model parameters are learnt with standard approach to resolve the ridge regression optimization:

$$\text{Argmin} \frac{1}{m} \sum_{i=1}^m \left(y^{(i)} + (\theta |x^{(i)}|)^2 + \lambda \|\theta\|^2 \right) \quad (1)$$

Where, the regularization parameter $\lambda > 0$ is used for finding the weighted ridge regularization term depicted as $\lambda \|\theta\|^2$

The parameter vector θ is found with this optimization when lambda is equal to zero in which the mean square error is minimized among the desired outputs $y^{(i)}$ model predictions over the training samples $x^{(i)}$.

In order to prevent over fitting from the model holding small training data, a ridge penalty has been utilized. The convex is first elevated optimization problem and the closed form expression is confessed from its solution. This solution can be derived by reformulating the optimization problem through writing matrix of $X \in L^{m \times d}$ using m rows analogous to the distinct d dimensional row vectors $x^{(i)}$ hence $X(i, j) = x_j^{(i)}$, and column vector of $Y \in L^m$ having n entries equivalent to the training labels $y^{(i)}$. Using these representations, the equation (1) is correspondent to

$$\text{argmin} \frac{1}{n} \|Y - X\theta\|^2 + \lambda \|\theta\|^2 \quad (2)$$

At this time, the gradient of the objective function is taken and it is equated with zero. Here it is possible to realize the optimization (1) which decreases to resolving this linear equation system.

$$\left(\frac{1}{n} X^Q X + \lambda I \right) \theta = \frac{1}{n} X^Q Y \quad (3)$$

I signifies the $d \times d$ identity matrix.

Thus, in order to resolve the linear equation system of the form $A \theta = b$, ridge regression is reduced. Based on the training data (X, Y) , the coefficients $A = \frac{1}{n} X^Q X + \lambda I$ and $b = \frac{1}{n} X^Q Y$ works. The positive definite linear system could be resolved with known algorithm as the coefficient matrix A is positive definite through building ridge regression.

3.2. Gradient descent

Tracking of gradients of cost function is followed for minimizing the gradient decent function. It comprised with well-known derivatives and form of cost, so it is easy to discern the gradient from the specified point and it may travel in that direction. i.e. Downhill towards the least value. To minimize the error of a model over training data, the similar technique named as stochastic gradient descent can be utilized in machine learning. The systematic approach is performed under realizing the every single instance towards model at a time and the prediction was made for that training instance, after calculating error on it, the model is updated for reducing the subsequent prediction of error. Utilizing this procedure, the set of coefficients present in model were found, hence model resultant in least error over training data.

The coefficient named weights are updated in machine learning language for every iteration using below equation Eqn 4

$$w_e = w_e - \text{alpha} * \text{delta} \quad (4)$$

w_e represents optimized weight or coefficient, the model error over the training data attributed to the weight is the gradient, alpha signifies rate of learning that must be organized (e.g. 0.1).

3.3. Compact thermal model

This model comprises with the layer assignment and thermal TSV effects. It is discussed below

(i) Impact of layer assignment

The compact thermal mode is utilized for calculating the power

Table 1

Description for Notations.

Notation	Description
T	temperature rise of the chip
d_l	Average Power dissipation of the layer l
$l_{\max}$	Number of layers
$l(t)$	total wire length of 3D IC
$l(e_l)$	wire length of m th layer
∞	temperature rise reduction quantity for each removal path
tsv_num	heat removal of paths
e_t	set of cells of m th layer and TSV which connected to m th layer
W_y	Layer y thermal resistance
W_z	Substrate thermal resistance

consumption of the resources which is the significant influence of increase in thermal rate. All-inclusive 3D IC is considered in the form of single tire stack.

It is signified as in the Eqn 5 below

$$TM = \sum_{i=1}^{l_{\max}} G_{li} \left(\sum_{k=1}^l W_y + W_z \right) \quad (5)$$

In equ (1), TR_i represents thermal resistance of the layer l , p_l signifies average power dissipation of the layer l and N be the number of layers, TR_s be the thermal resistance of the substrate.

(i) Thermal TSVs Impact in thermal model

The reduction over temperature drop is estimated and it is linearly proportional towards heat removal paths designed by TSVs. It is characterized as in Eqn 6

$$TM = \sum_{i=1}^{l_{\max}} G_{li} \left(\sum_{k=1}^l W_y + W_z \right) - tsv_num \times \infty \quad (6)$$

From the equation (2), it shows that the tsv_num as number of the removal paths of heat and ∞ signifies the temperature drop lessening showed as Eqn 7

$$\infty \times tsv_num. \quad (7)$$

(i) Effect of wire length

The 3D IC over-all wire length is premeditated using Eqn 8,

$$L(v) = \sum_k^T = l(e_t) \quad (8)$$

Equation (3) $L(v)$ signifies 3D IC total wire length, e_t signifies the set of cells of m th layer and TSV which connected to m th layer $l(e_t)$ signifies wire length of m th layer as in Eqn 9,

$$L(e_t) = \max_{v_k, v_m \in e_t} |x_k - x_m| + \max_{k_k, v_m \in e_t} |y_k - y_m| \quad (9)$$

TSV number of a net is calculated by using Eqn 10

$$v(e) = \max_{v_k, v_m \in e_t} |z_k - z_m| \quad (10)$$

	1	2	3	4
Layer1	a2			
Layer2	p3	a3		
Layer3	a1	a0	p1	p2

Fig. 3. Instance 1.

	1	2	3	4	5
Layer1	a2				
Layer2	a3				
Layer3	a1	p3	p2	a0	p1

Fig. 4. Instance 2.

	1	2	3	4
Layer1	a2			
Layer2	p3	a3		
Layer3	a1	a0	p1	p2

Fig. 5. Case 2.

The notations utilized in this work is described in below Table 1

An example concerning this work is shown below. Assume the $10\mu m^2$ for area of adder, $40\mu m^2$ for area of divisor, is $10\mu m^2$ for area of thermal TSV and the multiplier area is $20\mu m^2$. Likewise, the power dissipation of an adder is $10\mu W$, of divisor is $40\mu W$, multiplier is $30\mu W$. Every layer has the thermal resistance of around $0.1^\circ C/\mu W$, and substrate is $0.2^\circ C/\mu W$. The limit of footprint area is $100\mu m^2$. For each heat removal path, the quantity reduction of temperature upsurge is $5^\circ C$.

The comparison is made with three cases and it is given below

Instance 1:

Fig. 3 shows the layer assignment.

The rise in temperature is intended as

$$[(30 + 30 + 40) * (0.1 + 0.1 + 0.2)] + [(20 + 10 + 20) * (0.1 + 0.2)] + [10 * 0.2] = 57^\circ C$$

In layer 3, maximum device area is $80\mu m^2$. Addition of one heat removal path leads to the reduction in temperature rise of about $5^\circ C$. Therefore the temperature rise quantity is $51^\circ C$

Instance 2:

Fig. 4 layer assignment is given as

The increase in temperature can be calculated as,

$$[(10) * (0.1 + 0.1 + 0.2)] + [(20 + 10) * (0.1 + 0.2)] + [(40 + 20 + 30 + 20) * 0.2] = 36^\circ C$$

In layer 1, maximum device area is $100\mu m^2$. If there is no addition of heat path removal, the reduction in temperature rise of about $0^\circ C$. Therefore the temperature rise quantity is $38^\circ C$

Instance 3:

Fig. 5 layer assignment is shown as

The temperature rise can be calculated as,

$$[(10) * (0.1 + 0.1 + 0.2)] + [(20 + 10 + 10 + 30) * (0.1 + 0.2)] + [(30 + 40) * 0.2] = 39^\circ C$$

Table 2

Wire length Estimation (μm).

Benchmark	Stochastic Genetic Algorithm	Basic Genetic Algorithm	Linear Regression Algorithm
IBM 01	1.13E+06	1.15E+06	1.10E+06
IBM 02	1.15E+06	1.17E+06	1.13E+06
IBM 03	1.09E+06	1.13E+06	1.05E+06
IBM 04	1.17E+06	1.20E+05	1.12E+06
IBM 05	1.17E+06	1.19E+05	1.13E+06
IBM 06	1.13E+06	1.15E+06	1.10E+06
IBM 07	1.14E+06	1.17E+05	1.11E+06
IBM 08	1.16E+06	1.18E+06	1.14E+06
IBM 09	1.15E+06	1.17E+06	1.13E+06
IBM 10	1.09E+06	1.10E+06	1.07E+06
IBM 11	1.07E+06	1.11E+06	1.03E+06
IBM 12	1.11E+06	1.15E+06	1.09E+06
IBM 13	1.12E+06	1.15E+06	1.09E+06
IBM 14	1.16E+06	1.20E+06	1.11E+06
IBM 15	1.18E+06	1.23E+06	1.13E+06
IBM 16	1.20E+06	1.27E+06	1.17E+06
IBM 17	1.12E+06	1.16E+06	1.07E+06
IBM 18	1.14E+06	1.17E+06	1.11E+06

In layer 2, maximum device area is $60\mu m^2$. If two heat removal path is added, the reduction in temperature rise of around $10^\circ C$. Therefore the temperature rise quantity is $29^\circ C$

A routing approach is done after partitioning and the HVA model routing i.e. horizontal, vertical, and adjacent is conducted with the wirelength calculated using Eqn 11 Eqn 12 and the steps are shown below:

- Ø The grid measurement makes realistic solution through the availability of consent wires is described under the deliberation of wire width.
- Ø As the routing guidelines are determined at every layer rather consigned randomly, the complicated path is dropped utilizing fewer bias.
- Ø The capacitance on the wire occurs and generated from electrically powered association The released chip capacitance is isolated with the substrate even it persist among the single chip containing metal layers.
- Ø The horizontal and vertical wire cross over the adjacent films are the place where the capacitance occurs merely in the HVH configuration, it is less significant than the random routing.
- Ø The congestion is also reduced in this HVH configuration. The TSV information of vertical path is neglected by the straight association via Euclidean distance. Huge congestion may occur among the networks with the least possible association caused by random routing. This HSH channel yields higher ease to accomplish best solution in terms of observing less preventable methods and disruptive of networks.

$$wl = \sum_{k=1}^{k_n} (|z_r - z_s| + |x_r - x_s| + w) \quad (11)$$

$$= |y_k - y_m| + |z_k - z_m| + k_n * w \quad (12)$$

K_n is the layer count by authorized Net completely, w signifies height among neighboring layers and r, s represents edges in the net.

```

Algorithm 1: Linear Regression Algorithm
 $f_n[bs_{par}, ft_{ns}] = lin_{rg\_ag}([pa_{na}, ce_{na}], so_{no}, so_{na}, \emptyset);$ 
 $x = no_{na};$ 
 $y = \emptyset;$ 
 $th0_v = lin_{sp};$ 
 $th1_v = lin_{sp};$ 
 $aa = 1;$ 
for i = 1: 1  $th0_v$ 
    for j = 1: 1(  $th1_v$ )
         $[the_{ta1}] = gr_{co}^{ds}(x, the_{ta}, s_{na});$  %gradient_Descent_computation
     $[cl_{la}, cl_{so}] = ft_{fc}(x, the_{ta}, so_{no}, y);$ 
         $bs_{par}(aa) = cl_{la};$ 
         $ft_{ns} = cl_{so};$ 
    End
End
%gradient_Descent_computation

 $f_n[the_{ta1}] = gr_{co}^{ds}(x, the_{ta}, s_{na})$ 
global  $sl_{no}$ 
 $pr_1 = [];$ 
 $pr_2 = [];$ 
 $pr_3 = [];$ 
While  $ls_{em}(x_{nn})$ 
    If  $randi() == the_{ta1}$ 
         $s_l = randi(1) = the_{ta1}(x_{nn});$ 
         $sl_{no} = x_{nn} == s_l);$ 
    If  $sum(str_{cmp}(s_n, s_{na}));$ 
         $pr_1 = [pr_1, s_n];$ 
         $x_{nn}(x_{nn} == s_n);$ 
    Else
         $pr_2 = [pr_2, s_n];$ 
         $x_{nn}(x_{nn} == s_n);$ 
    End
    Elseif  $randi == the_{ta1}$ 
         $s_l = randi(1 \ 1(x_{nn}));$ 
         $s_l = x_{nn}(s_l);$ 
    If  $sum(str_{cmp}(s_n, s_{na}));$ 
         $pr_2 = [pr_2, s_n];$ 
         $x_{nn}(x_{nn} == s_n);$ 
    Else
         $pr_3 = [pr_3, s_n];$ 
         $x_{nn}(x_{nn} == s_n);$ 
    End
    Else
         $s_l = randi(1 \ 1(x_{nn}));$ 
         $s_n = x_{nn}(s_l);$ 
         $pr_3 = [pr_3, s_n];$ 
         $x_{nn}(x_{nn} == s_n);$ 
    End
End
End
 $the_{ta1}(1) = pr_1;$ 
 $the_{ta1}(2) = pr_2;$ 
 $the_{ta1}(3) = pr_3;$ 
% FitnessFunction
 $f_n[pa_{rt}, ft_{ns}] = ft_{fc}(no_{rp}, x, y, a)$ 

 $so_{ge} = y;$ 
For ii = 1: 1( $so_{ge}$ )
     $j_{va} = 0;$ 
     $sn_{se} = so_{ge}(ii);$ 
     $s_n = sn_{se};$ 

```

(continued on next page)

(continued)

```

For pp = 1:I(x);
     $l_y = x(pp);$ 
    If sum( $str_{cmp}(l_y, s_n)$ )
         $sn_{ly} = pp;$ 
    End
End
For jj = 2:I( $sn_{se}$ )
For pp = 1:I(x);
     $l_y = x(pp);$ 
    If sum( $str_{cmp}(l_y, s_n(jj))$ )
         $sn_{ly} = pp;$ 
    End
End
End
If  $sn_{ly} = sk_{ly}$ 
     $j_{va} = j_{va} + 0;$ 
Else
     $j_{va} = j_{va} + \text{abs}(sn_{ly} - sk_{ly});$ 
End
End
 $a_{data} = a;$ 
 $pr_{on} = x;$ 
 $no_{na} = no_{rp};$ 
 $sr_{ce} = y;$ 
A=rand(1 I( $no_{na}$ ));
B=rand(1 I( $no_{na}$ ));
 $wi_{le} = [];$ 
For ii = 1:I( $sr_{ce}$ )
    Con =  $sr_{ce}(ii);$ 
    If sum( $str_{cmp}(no_{na}, con)$ )==1
        Val =  $str_{cmp}(no_{na}, con)$ 
    end
For kk = 1:I(con)
    If sum( $str_{cmp}(no_{na}, con(kk))$ )==1
End
 $wi_{le} = (wi_{le} \text{ } wl);$ 
End
For ii = 1:I( $pr_{on}$ )
    Con1 = ( $pr_{on}(ii);$ 
If  $is_{em}(con1)$ 
    For kk = 1:I(con1)
        If sum( $str_{cmp}(con(kk), a_{data})$ )==1
    End
End
End
Con1=[];
End
 $r_f = 0.1;$ 
 $r_b = 0.2;$ 
 $pw_{ds} = \text{randi}(le_{th} (a_{ly}));$ 
 $tr_{ml} = \text{sum}(pw_{ds} * \text{sum}(r_{va}));$ 
 $ft_{rs} = [\text{sum}(wi_{le}) \text{sum}(tr_{ml}) \text{max}(a_{ly})];$ 

```

Table 3
Temperature Estimation (°C).

Benchmark	Stochastic Genetic Algorithm	Linear Regression Algorithm
IBM 01	3.50E+01	3.41E+01
IBM 02	3.62E+01	3.56E+01
IBM 03	3.74E+01	3.67E+01
IBM 04	3.86E+01	3.79E+01
IBM 05	3.98E+01	3.93E+01
IBM 06	4.10E+01	4.04E+01
IBM 07	4.22E+01	4.17E+01
IBM 08	4.34E+01	4.29E+01
IBM 09	4.46E+01	4.41E+01
IBM 10	4.58E+01	4.53E+01
IBM 11	4.70E+01	4.67E+01
IBM 12	4.82E+01	4.79E+01
IBM 13	4.94E+01	4.91E+01
IBM 14	5.06E+01	5.03E+01
IBM 15	5.13E+01	5.07E+01
IBM 16	5.19E+01	5.07E+01
IBM 17	5.25E+01	5.07E+01
IBM 18	5.31E+01	5.07E+01

Table 4
wire length (μm) comparison of the proposed approach.

Problem	Linear Regression Algorithm	Stochastic Genetic Algorithm	MRF routing	Box Router 2.0	GT-router
ibm01	61,256	63,368	64,327	63,531	66,953
ibm02	162,478	171,524	171,783	172,382	175,164
ibm03	147,167	146,918	146,991	147,492	149,076
ibm04	167,854	168,976	169,986	168,257	174,419
ibm05	406,963	408,991	409,764	411,138	417,756
ibm06	279,014	279,710	279,919	279,866	286,025
ibm07	368,264	368,674	368,520	368,002	373,977
ibm08	405,365	405,769	405,830	406,897	414,076
ibm09	413,478	415,215	414,789	417,303	421,702
ibm10	583,035	583,637	583,426	587,873	593,544

Table 5
Wire-length (μm) results on the ISPD2008 benchmarks.

Problem	Linear Regression Algorithm	Stochastic Genetic Algorithm	MRF routing	Box Router 2.0
adaptec1	27.6	35.24	51.48	58.37
adaptec2	16.25	30.4	48.22	55.69
adaptec3	93.79	101.29	128.05	137.96
adaptec4	78.64	88.54	123.22	127.79
adaptec5	87.50	105.71	145.71	162.11
Newblue1	28.16	43.42	62.23	78.88

4. Experimental analysis, verification

4.1. Dataset description

The ISPD98 Circuit Benchmark Suite:

For the training with Design Mechanization community, the standard sets are presented and retained by MCNC. However, hitches in partitioning and placement have not been adapted since five years. Then, in current standard public set the second major path consumes around 25,000 cells which is dignified minor via today's standards [20].

Improved Benchmarks - The old criteria having the number of divergences concerning networks and pins count. The dataset link is given: [21]

Old Benchmarks - These benchmarks take inconsistencies in terms of number of networks and pins [21]. Table 1 Represents the Descriptions for the Notations which is used in the algorithm and Expressions.

Table 2 and Table 3 depicts the estimation of wire length and the temperature from the benchmark starting from the IBM01 to IBM18 where the test problems varies

Table 4 and Figure 6 depicts the comparison of proposed wire length with traditional approaches. The linear regression achieved lesser wire length than the traditional approaches named as MRF routing [8], Box router 2.0 [22] and GT-router [23] and Starting from ibm01 to ibm10, the test problem differs

The Table 5 and Fig. 7 delivered here displays the ISPD2008 benchmarks wire length results. The test problem varies from adaptec1 to adaptec5. The proposed ELG is compared with the existing

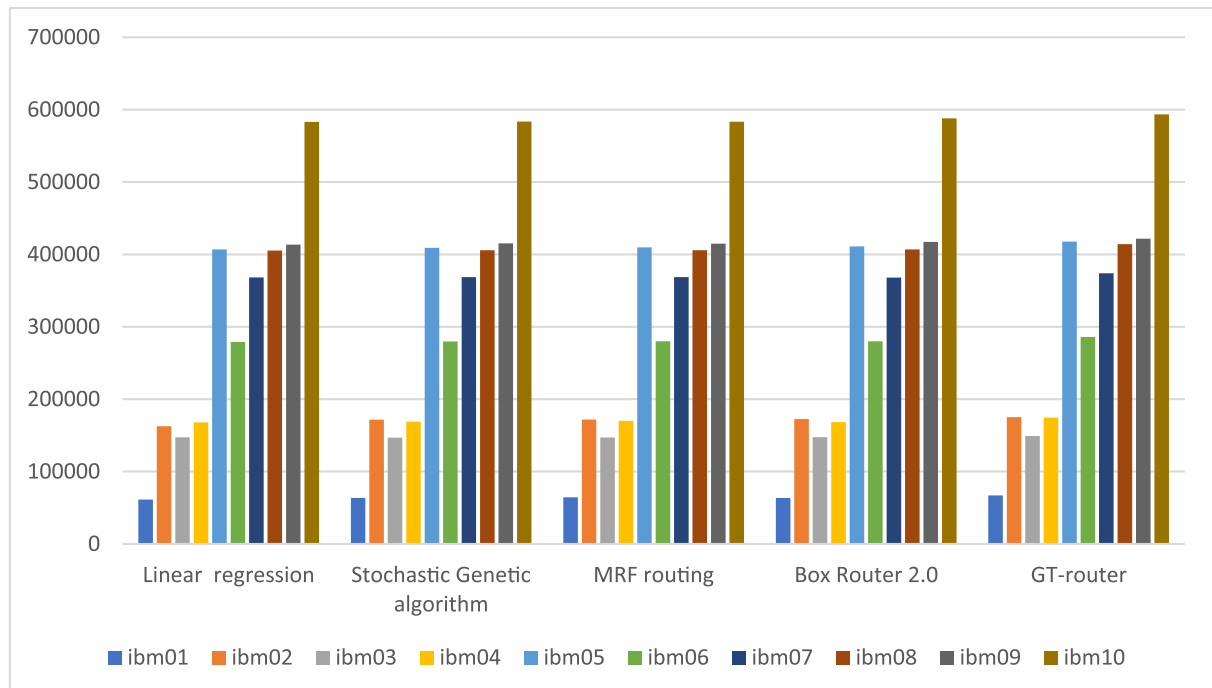


Fig. 6. Wirelength comparison.

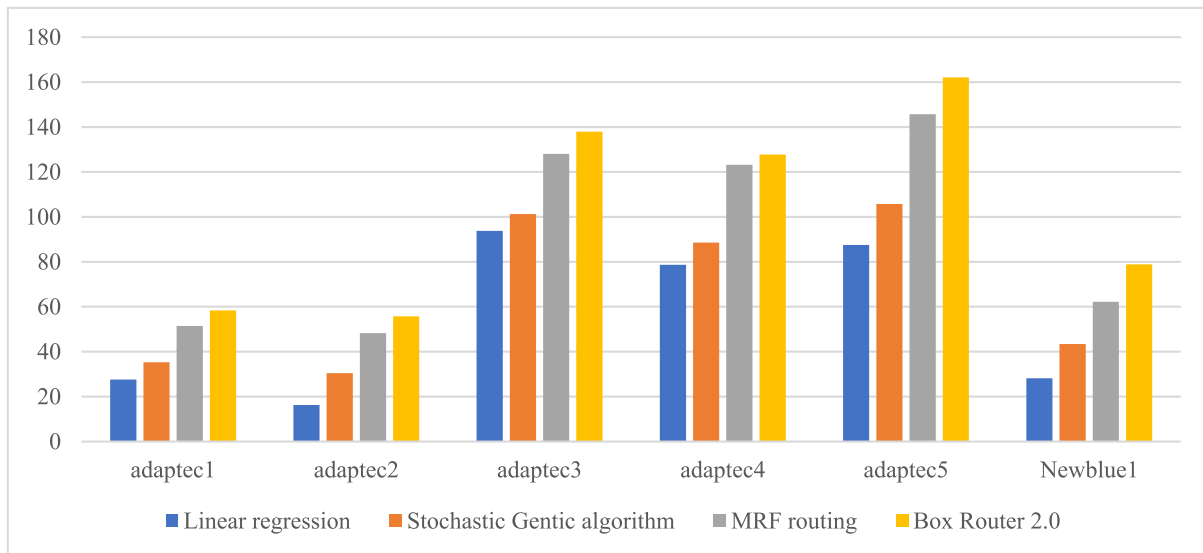


Fig. 7. ISPD2008 benchmarks wire length result.

Table 6

The ISPD2008 test problems.

Test problem	# of nets	Grid size
adaptec1	219,794	324 × 324
adaptec2	260,159	424 × 424
adaptec3	466,295	774 × 779
adaptec4	515,304	465 × 468
adaptec5	867,441	399 × 399
newblue1	331,663	557 × 463

techniques.

ISPD'98 test problems are shown in Table 6 which in turn displays the size of grid and # of nets and the test problem varies from ibm01 to ibm10.

5. Conclusion

It is concluded that the effective linear regression based layer assignment approach was achieved with the specific characteristics (i.e. gradient descent and fitness function) of linear regression algorithm. From the analysis of data it was revealed that the proposed ELRM is superior to other traditional approaches. This first top notch ML based approach towards layer assignment was accomplished in terms of higher reduction of temperature and wire length. The comparison was also made in order to prove the efficiency of proposed ELRM and it was accomplished as well. An equal efficient traditional approaches were compared to prove the proposed proficiency.

Declaration of Competing Interest

Please check the following as appropriate:

- All authors have participated in (a) conception and design, or analysis and interpretation of the data; (b) drafting the article or revising it critically for important intellectual content; and (c) approval of the final version.
- This manuscript has not been submitted to, nor is under review at, another journal or other publishing venue.
- The authors have no affiliation with any organization with a direct or indirect financial interest in the subject matter discussed in the manuscript.

References

- [1] K. Pandiaraj, P. Sivakumar, R. Sridevi, Minimization of wirelength in 3d IC routing by using differential evolution algorithm, in: 2017 IEEE International Conference on Electrical, Instrumentation and Communication Engineering (ICEICE), 2017, pp. 1–5, <https://doi.org/10.1109/ICEICE.2017.8191950>.
- [2] K. Pandiaraj, P. Sivakumar, N. Geetharamani, Reduction of temperature rise in 3D IC routing, in: 2017 IEEE International Conference on Electrical, Instrumentation and Communication Engineering (ICEICE), 2017, pp. 1–5, <https://doi.org/10.1109/ICEICE.2017.8191907>.
- [3] P. Sivakumar, K. Pandiaraj, K. Jeyaprasanth, Optimization of Thermal Aware Multilevel Routing For 3D IC, *IntJ Analog Integr Circ Signal Process* 103 (1) (October 2019), <https://doi.org/10.1007/s10470-019-01513-y>.
- [4] F. Ye, K. Chakrabarty, TSV open defects in 3D integrated circuits: characterization, test, and optimal spare allocation, in: DAC Design Automation Conference, 2012, pp. 1024–1030, <https://doi.org/10.1145/2228360.2228545>.
- [5] K. Athikulwongse, M. Ekpanyapong, S.K. Lim, Exploiting die-to-die thermal coupling in 3-D IC placement, *IEEE Trans. Very Large Scale Integr. VLSI Syst.* 22 (2013) 2145–2155, <https://doi.org/10.1145/2228360.2228495>.
- [6] D.H. Kim, K. Athikulwongse, S.K. Lim, Study of through-silicon-via impact on the 3-D stacked IC layout, *IEEE Trans. Very Large Scale Integr. VLSI Syst.* 21 (2012) 862–874, <https://doi.org/10.1109/TVLSI.2012.2201760>.
- [7] Y. Cheng, A. Todri-Sanial, J. Yang, W. Zhao, Alleviating Through-Silicon-Via Electromigration for 3-D Integrated Circuits Taking Advantage of Self-Healing Effect, *IEEE Trans. Very Large Scale Integr. VLSI Syst.* 24 (2016) 3310–3322, <https://doi.org/10.1109/TVLSI.2016.2543260>.
- [8] S.M. Sait, U.F. Siddiqi, A stochastic evolution algorithm based 2D VLSI global router, *Integr. VLSI J* 53 (2016) 115–125, <https://doi.org/10.1016/j.vlsi.2015.12.007>.
- [9] Y.-G. Chen, W.-Y. Wen, Y. Shi, W.-K. Hon, S.-C. Chang, Novel spare TSV deployment for 3-D ICs considering yield and timing constraints, *IEEE Trans. Comput. Aided Des. Integr. Circuits Syst.* 34 (2014) 577–588, <https://doi.org/10.1109/TCAD.2014.2385759>.
- [10] S. Pasricha, A framework for TSV serialization-aware synthesis of application specific 3D networks-on-chip, in: 2012 25th International Conference on VLSI Design, 2012, pp. 268–273, <https://doi.org/10.1109/VLSID.2012.82>.
- [11] T. Lu, C. Serafy, Z. Yang, S.K. Samal, S.K. Lim, A. Srivastava, TSV-based 3-D ICs: design methods and tools, *IEEE Trans. Comput. Aided Des. Integr. Circuits Syst.* 36 (2017) 1593–1619, <https://doi.org/10.1109/TCAD.2017.2666604>.
- [12] Zou Q., Zhang T., Kursun E., Xie Y., Thermomechanical stress-aware management for 3D IC designs, *Proceedings of the Conference on Design, Automation and Test in Europe*, 2013, pp. 1255–1258, <https://doi.org/10.7873/DAT.2013.260>.
- [13] X. Zhao, M.R. Scheuermann, S.K. Lim, Analysis and modeling of DC current crowding for TSV-based 3-D connections and power integrity, *IEEE Trans. Compon., Packaging and Manufac. Technol.* 4 (2013) 123–133, <https://doi.org/10.1109/TCPMT.2013.2276779>.
- [14] M.B. Healy, S.K. Lim, Distributed TSV topology for 3-D power-supply networks, *IEEE Trans. Very Large Scale Integr. VLSI Syst.* 20 (2011) 2066–2079, <https://doi.org/10.1109/TVLSI.2011.2167359>.
- [15] M.M. Shulaker, T.F. Wu, A. Pal, L. Zhao, Y. Nishi, K. Saraswat, et al., Monolithic 3D integration of logic and memory: carbon nanotube FETs, resistive RAM, and silicon FETs, in: 2014 IEEE International Electron Devices Meeting, 2014, <https://doi.org/10.1109/IEDM.2014.7047120>, 27.4. 1–27.4. 4.
- [16] M. Lin, A.El Gamal, Y.-C. Lu, S. Wong, Performance benefits of monolithically stacked 3-D FPGA, *IEEE Trans. Comput. Aided Des. Integr. Circuits Syst.* 26 (2007) 216–229, <https://doi.org/10.1109/TCAD.2006.887920>.

- [17] Das S., Doppa J.R., Kim D.H., Pande P.P., Chakrabarty K., Optimizing 3D NoC design for energy efficiency: a machine learning approach, Proceedings of the IEEE/ACM International Conference on Computer-Aided Design, 2015, pp. 705–712. <https://doi.org/10.1109/ICCAD.2015.7372639>.
- [18] Y.-J. Huang, C.-L. Pan, S.-C. Lin, M.-H. Guo, Machine-learning approach in detection and classification for defects in TSV-based 3-D IC, IEEE Trans. Compon., Packag. Manuf. Technol. 8 (2018) 699–706, <https://doi.org/10.1109/TCPMT.2017.2788896>.
- [19] H. Sung, K. Cho, K. Yoon, S. Kang, A delay test architecture for TSV with resistive open defects in 3-D stacked memories, IEEE Trans. Very Large Scale Integr. VLSI Syst. 22 (2013) 2380–2387, <https://doi.org/10.1109/TVLSI.2013.2289964>.
- [20] T. I. C. B. Suite. <https://vlsicad.ucsd.edu/UCLAWeb/cheese/ispd98.html>.
- [21] I. benchmarks. <http://www.ispd.cc/contests/08/ispd08rc.html>.
- [22] M. Cho, K. Lu, K. Yuan, D.Z. Pan, BoxRouter 2.0: a hybrid and robust global router with layer assignment for routability, ACM Trans. Desig. Automat. Electro. Syst. (TODAES) 14 (2009) 32, <https://doi.org/10.1145/1497561.1497575>.
- [23] U.F. Siddiqi, S.M. Sait, Y. Shiraishi, A game theory-based heuristic for the two-dimensional VLSI global routing problem, J. Circ. Syst. Comput. 24 (2015), 1550082, <https://doi.org/10.1142/S0218126615500826>.
- [24] S. Wang, K. Chakrabarty, M.B. Tahoori, Defect clustering-aware spare-TSV allocation in 3D ICs for yield enhancement, IEEE Trans. Comput. Aided Des. Integr. Circuits Syst. (2018), <https://doi.org/10.1109/ICCAD.2015.7372585>.
- [25] K. Cao, J. Zhou, T. Wei, M. Chen, S. Hu, K. Li, A survey of optimization techniques for thermal-aware 3D processors, J. Syst. Archit. (2019), <https://doi.org/10.1016/j.sysarc.2019.01.003>.



K. Pandiaraj is currently an Assistant Professor in the Department of Electronics and Communication Engineering at Kalasalingam Academy of Research and Education (Kalasalingam University), Krishnankoil, India since 2008. He graduated in Electronics and Communication Engineering from Anna University and post graduated in VLSI Design in 2006 from Anna University, Chennai, India. He has published 6 technical research papers in reputed International Journals and IEEE Conferences and published 23 papers in International and national Level Conferences. He is doing his Ph.D. in the area of 3D IC Physical Design at Kalasalingam Academy of Research and Education.



P. Sivakumar is currently an Professor and Head in the Department of Electronics and Communication Engineering at Kalasalingam Academy of Research and Education (Kalasalingam University), Krishnankoil, India since 2008. I am graduated in Electronics and Communication Engineering from Anna University and post graduated in VLSI Design Sastra University, Thanjavur, India. I had 15 years' experience in Research and Development, Teaching. I has published 15 technical research papers in reputed National and International Journals. I had completed my Ph.D. in the area of VLSI Physical Design at Anna University. He is also interested in Machine Intelligence research.



K. JeyaPrakash is currently an Assistant Professor in the Department of Electronics and Communication Engineering at Kalasalingam Academy of Research and Education (Kalasalingam University), Krishnankoil, India since 2009. Prior to his appointment at Kalasalingam, he worked as a lecturer in Electronics and Communication Engineering department at RMK Engineering College, Kavarapettai from 2007 to 2009 and as a lecturer in Electronics and Communication Engineering Department at Sethu Institute of Technology, Puloor from 2003 to 2005. He completed his M.E. in VLSI Design from Anna University, Chennai and B.E. in Electronics and Communication Engineering from Bharathidasan University, Tiruchirappalli in the years 2007 and 2000 respectively. He is doing his Ph.D. in the area of 3D IC Physical Design at Kalasalingam University. He is also interested in implementation of Problem based Learning and Outcome Based Education in Engineering.